Variation in two dictionaries of Botlikh

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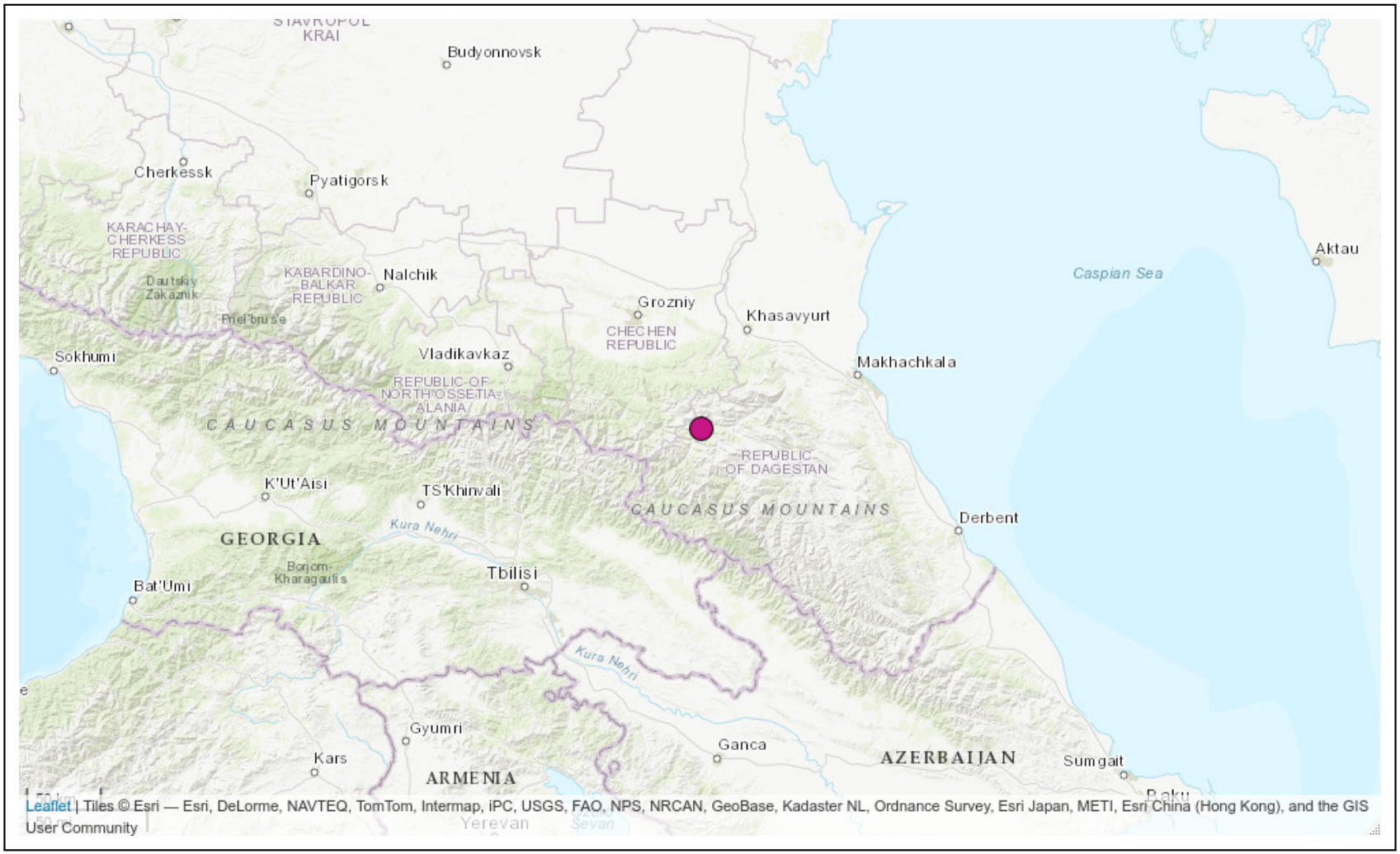
Botlikh

- Botlikh < Andic group < East Caucasian language family
- Unwritten (can be written with extended Cyrillic script for Avar)
- ~5000-8000 speakers
- Mostly spoken in 3 villages in northwestern Daghestan (Russian Federation): Botlikh, Miarso, Ashino, (Ankho); minor dialectal differences
- Opinions vary on the language's vitality — it is still passed on to children and spoken at home, but some families are shifting to Russian

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- Opinions vary on the language's vitality — it is still passed on to children and spoken at home, but some families are shifting to Russian
- One full reference grammar in Georgian (Gudava 1962)
- Two dictionaries:
  (Saidova & Abusov 2012) and (Alekseev & Azaev 2019)

Botlikh


STAVROPOL KRAI
Budyonnovsk
Cherkessk
Pyatigorsk
KARACHAY-CHERKESS REPUBLIC
Dautskiy Zakaznik
KABARDINO-BALKAR REPUBLIC
Nalchik
Priel'brus'e
Grozniy
CHECHEN REPUBLIC
Khasavyurt
Aktau
Caspian Sea
Sokhumi
Vladikavkaz
Makhachkala
REPUBLIC OF NORTH OSSETIA-ALANIA
CAUCASUS MOUNTAINS
REPUBLIC OF DAGESTAN
K'Ut'Aisi
TS'Khinvali
CAUCASUS MOUNTAINS
Derbent
Kura Nehri
GEORGIA
Borjom-Kharagauli's
Tbilisi
Bat'Umi
Kura Nehri
Gyumri
Ganca
AZERBAIJAN
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Two dictionaries

(Saidova & Abusov 2012)

- Compiled in the 2000s by a native speaker (M.G. Abusov) and an experienced linguist (P.A. Saidova)
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(Alekseev & Azaev 2019)

- Compiled in the 1960s / 1970s by a native speaker / philologist (X.G. Azaev) and later (in the 2000s) systematized by an experienced linguist (M.E. Alekseev)
- Subsequently edited by T.A. Maisak and scheduled for posthumous publication later this year
- Botlikh only

Two dictionaries

- Dictionaries were compiled *independently* of each other
- ~8,000 headwords (Saidova & Abusov 2012) vs. ~9,000 'words and expressions' (Alekseev & Azaev 2019)
- No metadata on the speakers consulted
- Data from (Alekseev & Azaev 2019) was collected several decades earlier, but M.G. Abusov also consulted elderly speakers with the aim of collecting archaic vocabulary (p.c.)

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- At first glance, the two resources seemed to display variation

Two dictionaries

Aim of the present investigation

Compare the two resources, see to what extent they overlap and diverge, and verify whether they display systematic variation

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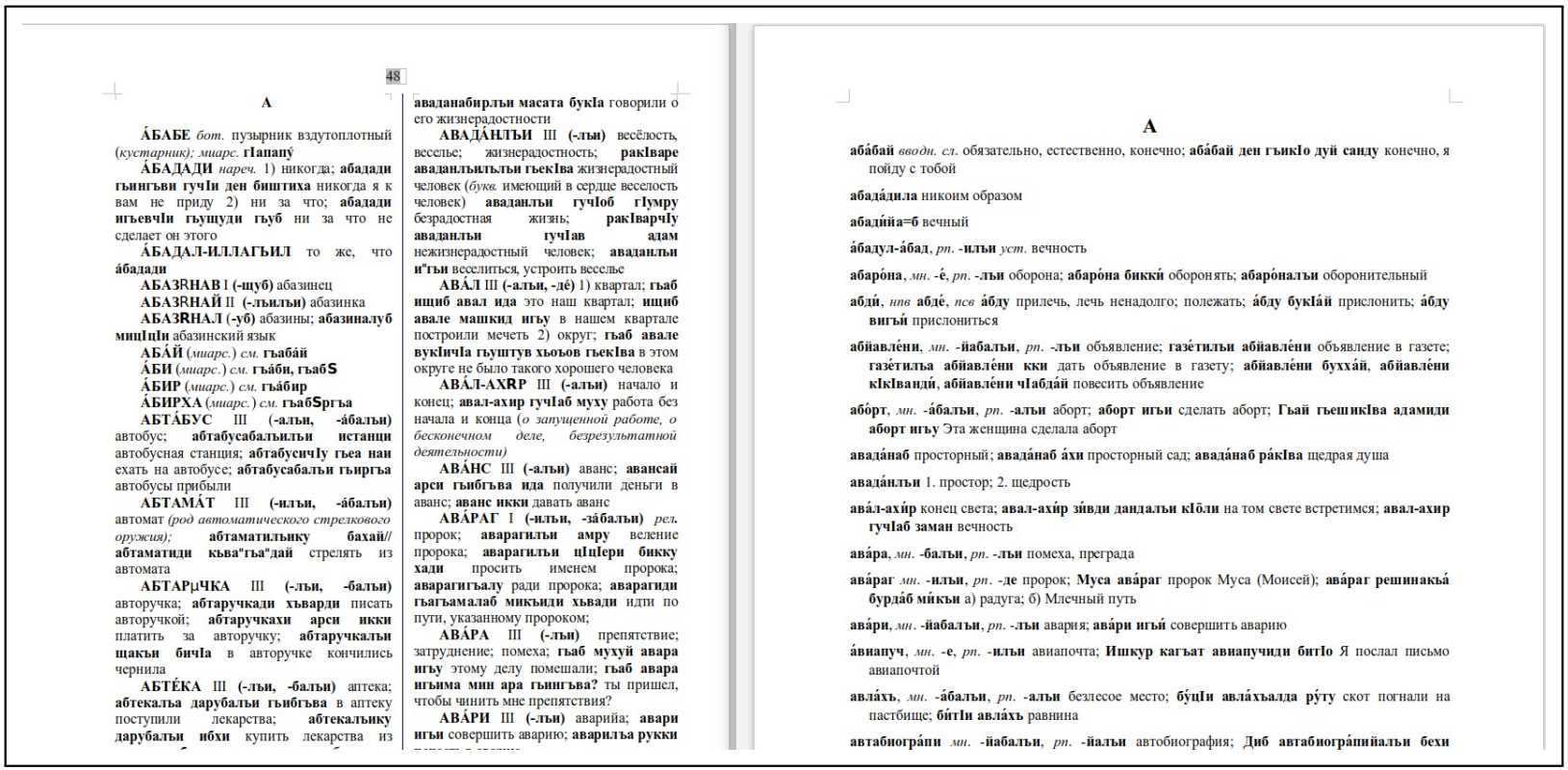
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...

and *hopefully*, explain the kind of variation we found (idiolectal, diachronic, etc.)

Data

Merge dictionaries (doc to xls table) through a painstaking process of unification (George Moroz).



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А

А́БАБЕ *бот.* пузырник вздутоплотный (*кустарник*); *миарс.* **гӀапапу́**

А́БАДАДИ *нареч.* 1) никогда; **абадади гьингъви гучӀи ден биштиха** никогда я к вам не приду 2) ни за что; **абадади игъевчӀи гьущуди гьуб** ни за что не сделает он этого

А́БАДАЛ-ИЛЛАГЬИЛ то же, что **а́бадади**

АБАЗ1НАВ I **(-щуб)** абазинец

АБАЗ1НАЙ II **(-лъилъи)** абазинка

АБАЗ1НАЛ (-уб) абазины; **абазиналуб мицӀцӀи** абазинский язык

АБА́Й (*миарс.*) *см.* **гъаба́й**

А́БИ (*миарс.*) *см.* **гъа́би, гъабS**

А́БИР (*миарс.*) *см.* **гъа́бир**

А́БИРХА (*миарс.*) *см.* **гъабSргъа**

АБТА́БУС III **(-алъи, -а́балъи)** автобус; **абтабусабалъилъи истанци** автобусная станция; **абтабусичӀу гьеа наи** ехать на автобусе; **абтабусабалъи гьиргъа** автобусы прибыли

АБТАМА́Т III **(-илъи, -а́балъи)** автомат *(род автоматического стрелкового оружия)*; **абтаматилъику бахай// абтаматиди кьва"гьа"дай** стрелять из автомата

АБТАРЦЧКА III **(-лъи, -балъи)** авторучка; **абтаручкади хъварди** писать авторучкой; **абтаручкахи арси икки** платить за авторучку; **абтаручкалъи щакъи бичӀа** в авторучке кончились чернила

АБТЕ́КА III **(-лъи, -балъи)** аптека; **абтекалъа дарубалъи гьибгъва** в аптеку поступили лекарства; **абтекалъику дарубалъи ибхи** купить лекарства из **аваданабирлъи масата букӀа** говорили о его жизнерадостности

АВАДА́НЛЪИ III **(-лъи)** весёлость, веселье; жизнерадостность; **ракӀваре аваданлъилъилъи гьекӀва** жизнерадостный человек (*букв.* имеющий в сердце веселость человек) **аваданлъи гучӀоб гӀумру** безрадостная жизнь; **ракӀварчӀу аваданлъи гучӀав адам** нежизнерадостный человек; **аваданлъи и"гъи** веселиться, устроить веселье

АВА́Л III **(-алъи, -дé)** 1) квартал; **гъаб ищиб авал ида** это наш квартал; **ищиб авале машкид игъу** в нашем квартале построили мечеть 2) округ; **гъаб авале вукӀичӀа гъуштув хьоъов гьекӀва** в этом округе не было такого хорошего человека

АВА́Л-АХ1Р III **(-алъи)** начало и конец; **авал-ахир гучӀаб муху** работа без начала и конца (*о запущенной работе, о бесконечном деле, безрезультатной деятельности*)

АВА́НС III **(-алъи)** аванс; **авансай арси гьибгъва ида** получили деньги в аванс; **аванс икки** давать аванс

АВА́РАГ I **(-илъи, -за́балъи)** *рел.* пророк; **аварагилъи амру** веление пророка; **аварагилъи цӀцӀери бикку хади** просить именем пророка; **аварагигъалу** ради пророка; **аварагиди гьагъамалаб микъиди хьвади** идти по пути, указанному пророком;

АВА́РА III **(-лъи)** препятствие; затруднение; помеха; **гъаб мухуй авара игъу** этому делу помешали; **гъаб авара игъима мин ара гьингъва?** ты пришел, чтобы чинить мне препятствия?

АВА́РИ III **(-лъи)** аварийа; **авари игъи** совершить аварию; **аварилъа рукки**

А

аба́бай *вводн. сл.* обязательно, естественно, конечно; **аба́бай ден гьикӀо дуй санду** конечно, я пойду с тобой

абада́дила никоим образом

абади́йа=б вечный

а́бадул-а́бад, *рп.* **-илъи** *уст.* вечность

абаро́на, *мн.* **-е́**, *рп.* **-лъи** оборона; **абаро́на бикки́** оборонять; **абаро́налъи** оборонительный

абди́, *нтв* **абде́**, *псв* **а́бду** прилечь, лечь ненадолго; полежать; **а́бду букӀа́й** прислонить; **а́бду вигъи́** прислониться

абйавле́ни, *мн.* **-йабалъи**, *рп.* **-лъи** объявление; **газе́тилъи абйавле́ни** объявление в газете; **газе́тилъа абйавле́ни кки** дать объявление в газету; **абйавле́ни кӀкӀвандй́, абйавле́ни чӀабда́й** повесить объявление

або́рт, *мн.* **-а́балъи**, *рп.* **-алъи** аборт; **аборт игъи** сделать аборт; **Гьай гьешикӀва адамиди аборт игъу** Эта женщина сделала аборт

авада́наб просторный; **авада́нав а́хи** просторный сад; **авада́нав ра́кӀва** щедрая душа

авада́нлъи 1. простор; 2. щедрость

ава́л-ахи́р конец света; **авал-ахи́р зи́вди дандалъи кӀо́ли** на том свете встретимся; **авал-ахир гучӀаб заман** вечность

ава́ра, *мн.* **-балъи**, *рп.* **-лъи** помеха, преграда

ава́раг *мн.* **-илъи**, *рп.* **-де** пророк; **Муса ава́раг** пророк Муса (Моисей); **ава́раг решинакьа́ бурда́б ми́кьи** а) радуга; б) Млечный путь

ава́ри, *мн.* **-йабалъи**, *рп.* **-лъи** авария; **ава́ри игъй** совершить аварию

а́виапуч, *мн.* **-е**, *рп.* **-илъи** авиапочта; **Ишкур кагъат авиапучиди битӀо** Я послал письмо авиапочтой

авла́хъ, *мн.* **-а́балъи**, *рп.* **-алъи** безлесое место; **бу́цӀи авла́хъалда ру́ту** скот погнали на пастбище; **би́тӀи авла́хъ** равнина

автабиогра́пи *мн.* **-йабалъи**, *рп.* **-йалъи** автобиография; **Диб автабиогра́пийалъи бехи**

Data

Nouns (genitive, plural) and verbs (habitual and aorist)

lemma	pos	noun_gen	noun_pl	verb_prs	verb_pst	reference
арбагIзи́в	noun	-лъи				Saidova, Abusov
арджа́н	noun	-алъи	-			Alekseev 2006
аре́нда	noun	-лъи				Saidova, Abusov
аржаж/а́й	verb			-аймале́	-о́	Saidova, Abusov
аржа́й	verb			-е́	-о	Alekseev 2006
арж/а́й	verb			-а́ймале	-о́	Saidova, Abusov
а́ржар	noun	-лъи	-де́			Alekseev 2006
аржи́	verb			-е́	-а	Alekseev 2006
а́рж/и	verb			-е	-о	Saidova, Abusov

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аржи́	verb			-е́	-а	Alekseev 2006
а́рж/и	verb			-е	-о	Saidova, Abusov

From ~15,000-17,000 lexical entries, we automatically extracted 7,583 nouns and verbs, further manual correction

Parameters

- Phonology: word similarity, stress patterns
- Nominal morphology: formation of genitive case and plural forms
- Verbal morphology: formation of basic tenses

Phonology

For phonological analysis we used only those words which were found in both dictionaries (1,688).

- 1,309 words are the same
- 254 words have difference in stress: *ansːí* 'warm up' (A&A) vs. *ánsːi* (S&A)
- and only 125 words have some segmental differences
  - Some differences could be typos (*kalχoznik'* 'kolkhoz member' (A&A) vs. *kalχoznik* (S&A), *lah* (лагь) 'slave' (A&A) 2006 vs. *laʁ* (лагъ) (S&A))
  - Multiple cases where there is a small difference that could be explained either as a typo or in terms phonological variation (*čuhí* 'to run' (A&A) vs. *čũhí* (S&A), *kusu* 'cherry plum' (A&A) vs. *kusːu* (S&A))
  - Multiple cases where Russian borrowings were adopted differently (*pojiz* 'train' (A&A) vs. *poezd* (S&A), *biton* 'milk can' (A&A) vs. *bitun* (S&A))

Nominal morphology

Formation of the plural

- A suffix is attached to the absolutive stem: *na* 'thing' < *na-**bałi*** 'things'
- With stems ending in a consonant the vowel *-a-* is often inserted before the suffix: *majmalak* 'monkey' < *majmalak-**a**-**bałi*** 'monkeys'
- With stems ending in a vowel alternation can occur: *ruš**a*** 'tree' < *ruš**i**-**bałi*** 'trees', *sal**u*** 'tooth' < *sal**a**-**bałi*** 'teeth', *burał**i*** 'pitcher' < *burał**a**-**bałi*** 'pitchers'
- Among the most common suffixes are: *-bałi* (and variants, e.g. *-zabałi*, *-małi*, etc.), *-de*, *-(w)e*

Nominal morphology

Case declension (core cases)

- I type — the stem does not change when a suffix is attached (mostly stems ending in a vowel and masdars): *babu* 'mom' < *babu-ɬi* (genitive), *masir* 'measurement' < *masir-ɬi* (genitive)
- II type — case suffixes are attached to the oblique stem of the noun (mostly stems ending in a consonant, sometimes stems ending in a vowel): *askar* 'army' < *askar-**a**-ɬi* (genitive), *din* 'religion' < *din-**i**-ɬi* (genitive), *ima* 'father' < *im**u**-ɬi* (genitive)

Nominal morphology

- Both dictionaries report both the **genitive** and the **plural** suffix
- We used this information to study the productivity of such suffixes, and to see whether the two dictionaries display variation

Nominal morphology

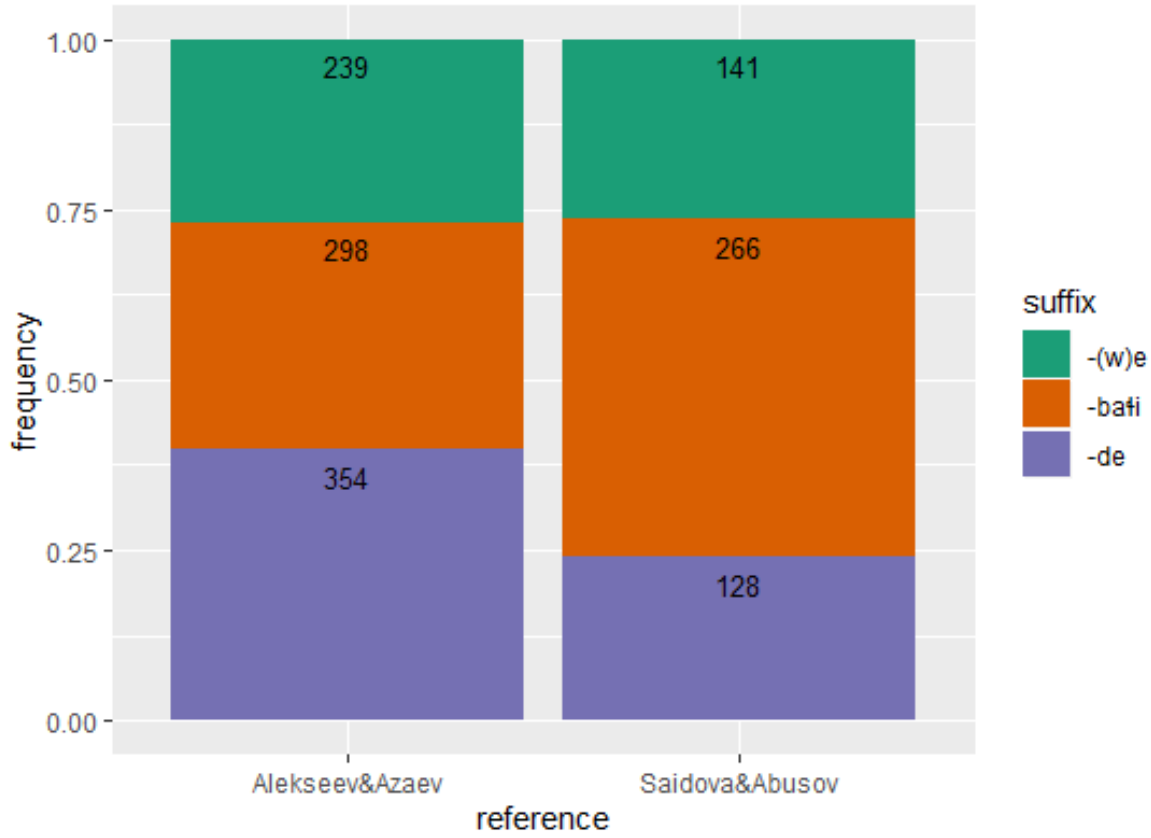
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- We retrieved 2,965 nouns for A&A, 2,952 for S&A
- 1,072 pairs were present in both dictionaries and had information about inflectional forms

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- 1,072 pairs were present in both dictionaries and had information about inflectional forms
- The genitive suffix is almost always reported in both dictionaries: for all 1,072 words in S&A, for 1,066 words in A&A
- The plural is not reported for all words: for 571 words in S&A, for 879 words in A&A

Nominal morphology

Figure 1: Variation: plural suffixes



(X-squared = 47.118, df = 2, p-value = 5.869e-11)

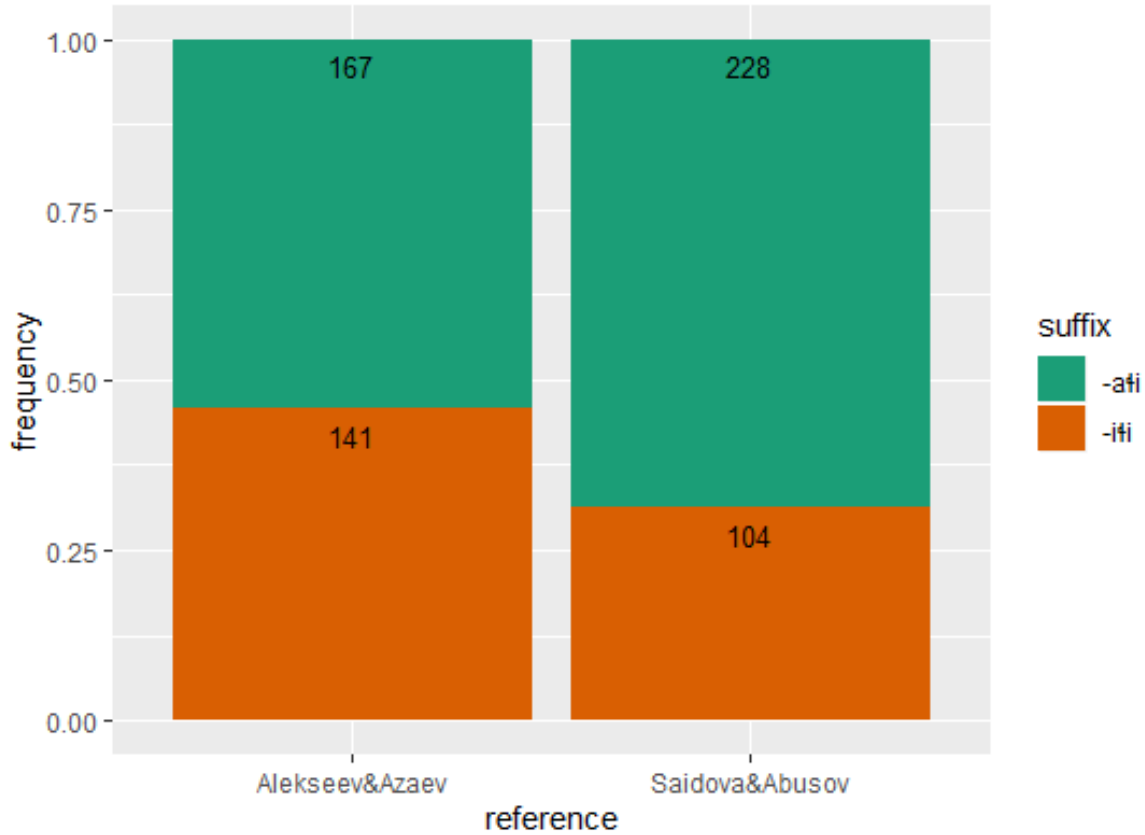
Nominal morphology

Table 1: Stem endings and genitive suffix

Suffix	*-ɬi*		*-aɬi*		*-iɬi*		*-uɬi*	
Stem	S&A	A&A	S&A	A&A	S&A	A&A	S&A	A&A
consonant	232	266	**228**	**167**	**104**	**141**	-	1
-a	182	167	-	-	3	13	15	14
-i	143	151	10	4	-	-	3	2
-u	81	78	1	-	-	3	-	-
-e	6	6	-	-	-	-	-	-
-o	7	7	-	-	-	-	-	-

Nominal morphology

Figure 2: Variation: genitive in *-aɬi* vs. *-iɬi* with stems ending in a consonant



(X-squared = 13.523, df = 1, p-value = 0.0002357)

Variation: examples

- It seems that variation frequently involves loanwords:
  - *dakument* 'document' < *dakument-**ałi*** (genitive, S&A) vs. *dakument-**iłi*** (genitive, A&A)
  - *birgadir* 'foreman' < *birgadir-**zabałi*** (plural, S&A) vs. *birgadir-**de*** (plural, A&A)
  - *kassir* 'cashier' < *kassir-**ałi*** (genitive, S&A), *kassir-**zabałi*** (plural, S&A) vs. *kassir-**iłi*** (genitive, A&A), *kassir-**de*** (plural, A&A)

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- But not only:
  - *gač'a* 'hoe' < *gač'a-**ɬi*** (genitive, S&A), *gač'-**ibałi*** (plural, S&A) vs. *gač'-**iłi*** (genitive, A&A), *gač'-**e*** (plural, A&A)

Verbal morphology

- We retrieved 1,570 verbal entries for A&A and 1,475 for S&A
- Only 554 pairs were present in both dictionaries and had information about inflectional forms

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- Infinitive and Aorist are indicative for conjugation:

Verbal morphology

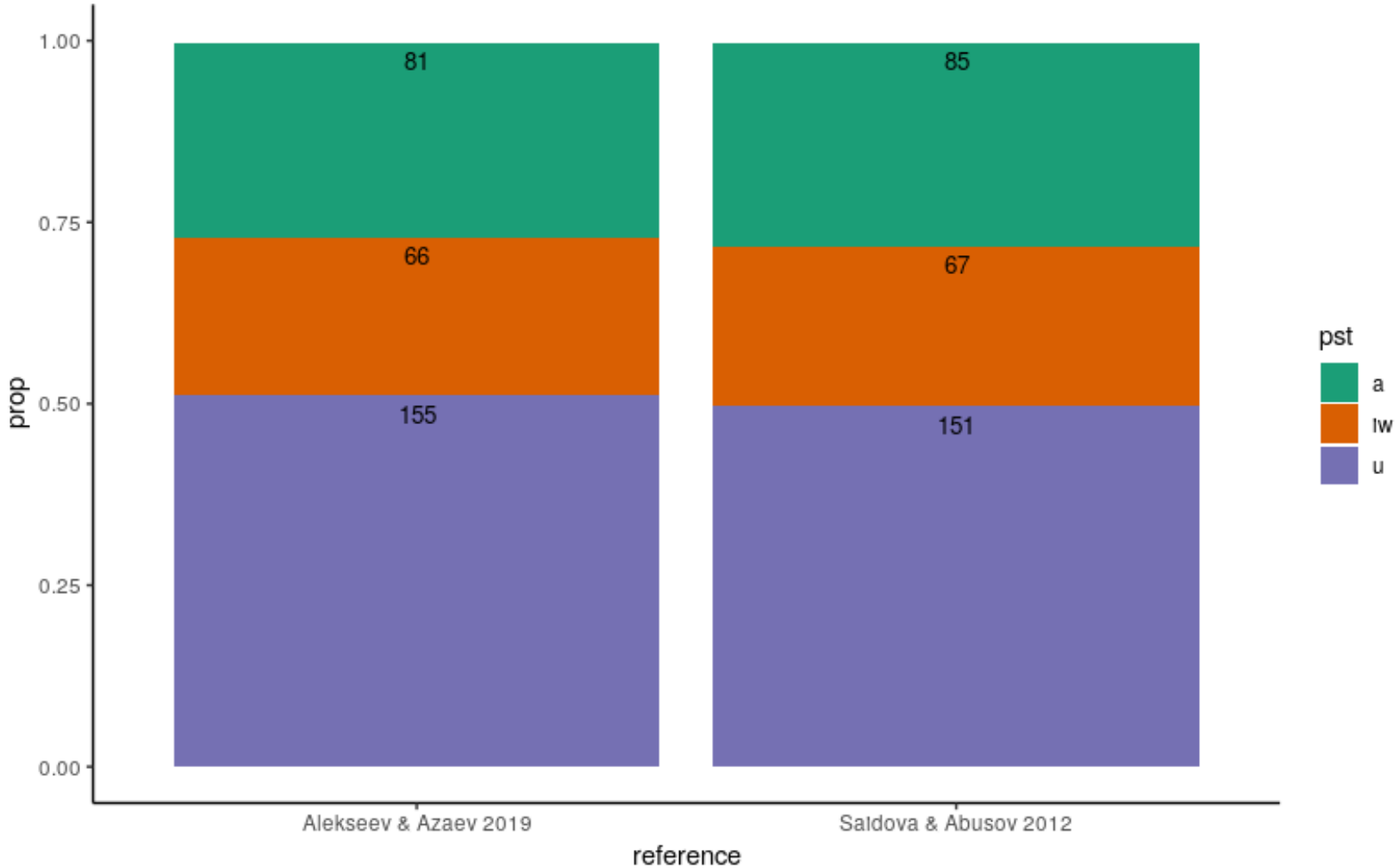
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Table 2: Basic verb inflection in Botlikh

	infinitive	present	past
Basic	-i	-e	-a / -u / -iw
Derived	-ɬi	b-ah-e	b-ah-u 'become'
		b-uk'-e	b-uk'-a 'be'
Causative	-a-j	mal-e / malih-e	-o / malih-u

Verbal morphology

Figure 3: Past tense suffixes in the sample of common verbs



Verbal morphology

- There is more variation than the plot shows

Table 3: Past suffixes in both dictionaries

S&A / A&A	a	iw	u	other
a	72	6	3	4
iw	3	55	6	3
u	3	4	142	2
other	3	1	4	0

Verbal morphology

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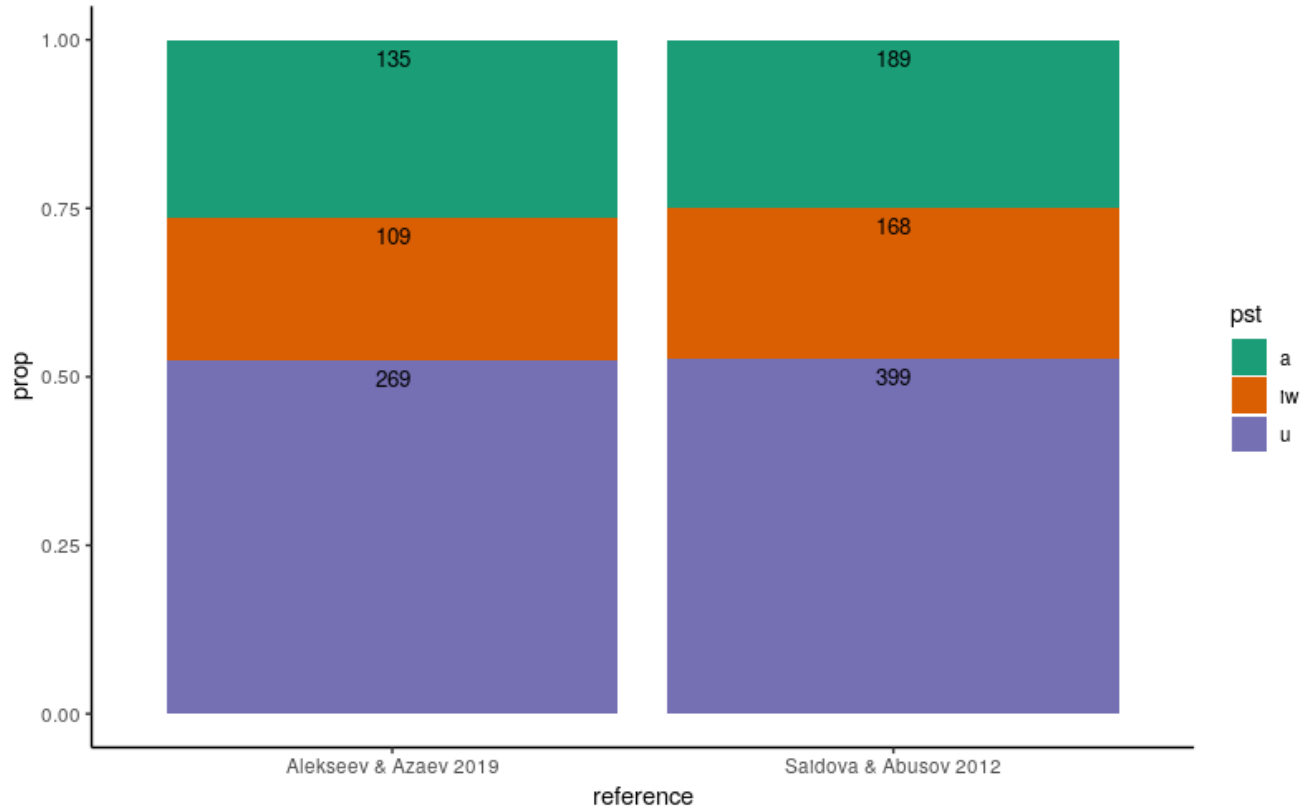
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other	3	1	4	0

- But still not a lot
- And no systematic correspondence

Verbal morphology

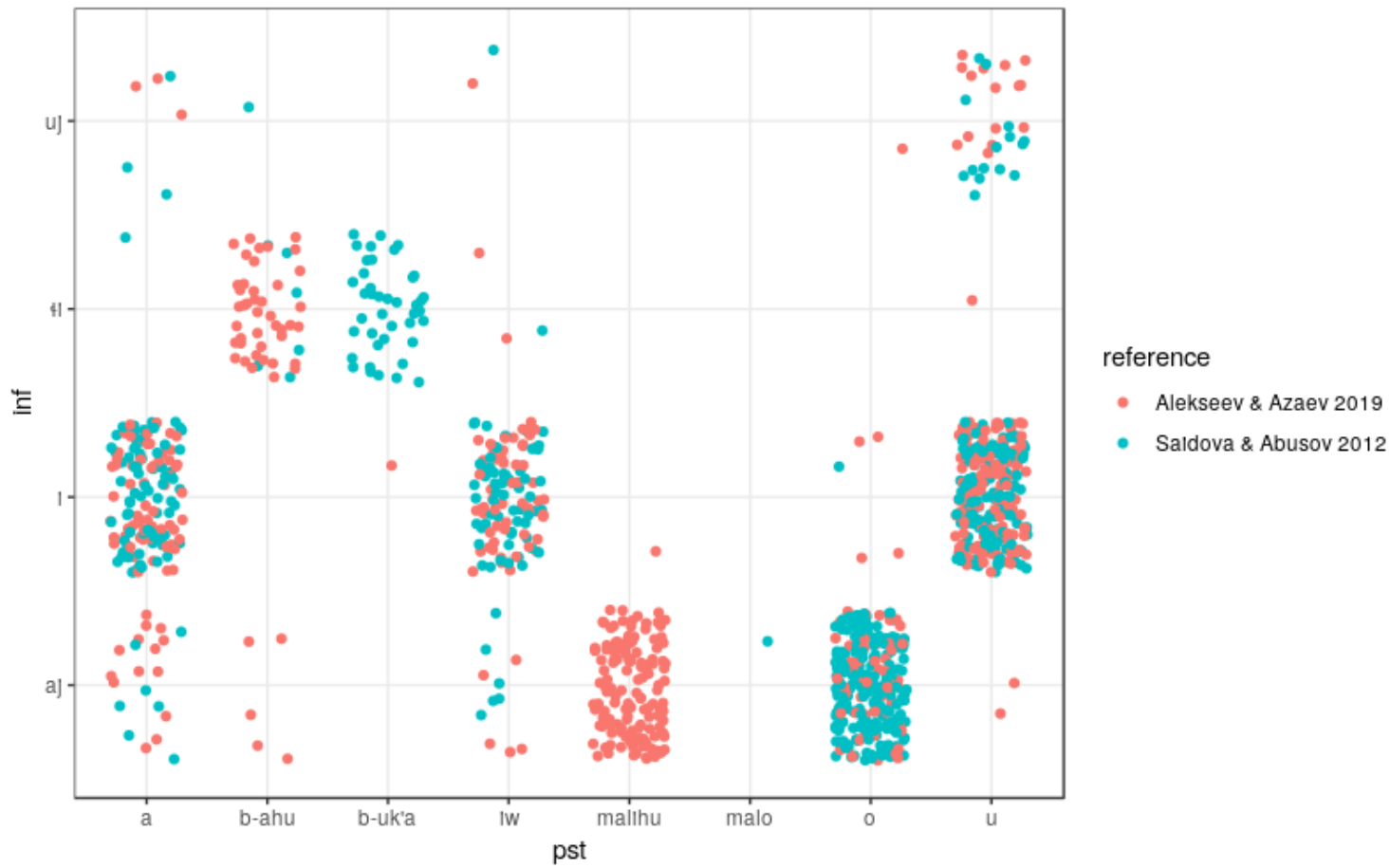
Figure 4: Past tenses in all dictionary data



(x-squared = 0.347, df = 2, p-value = 0.8407)

Verbal morphology

Figure 5: Infinitive vs. past tense



Verbal morphology

Table 4: Causative paradigms in Botlikh

	infinitive	present	past
A&A	-a-j	malih-e	malih-u
		-e	-o
S&A	-a-j	mal-e	-o
		-e	-o
		mal-e	mal-o (1x)

Verbal morphology

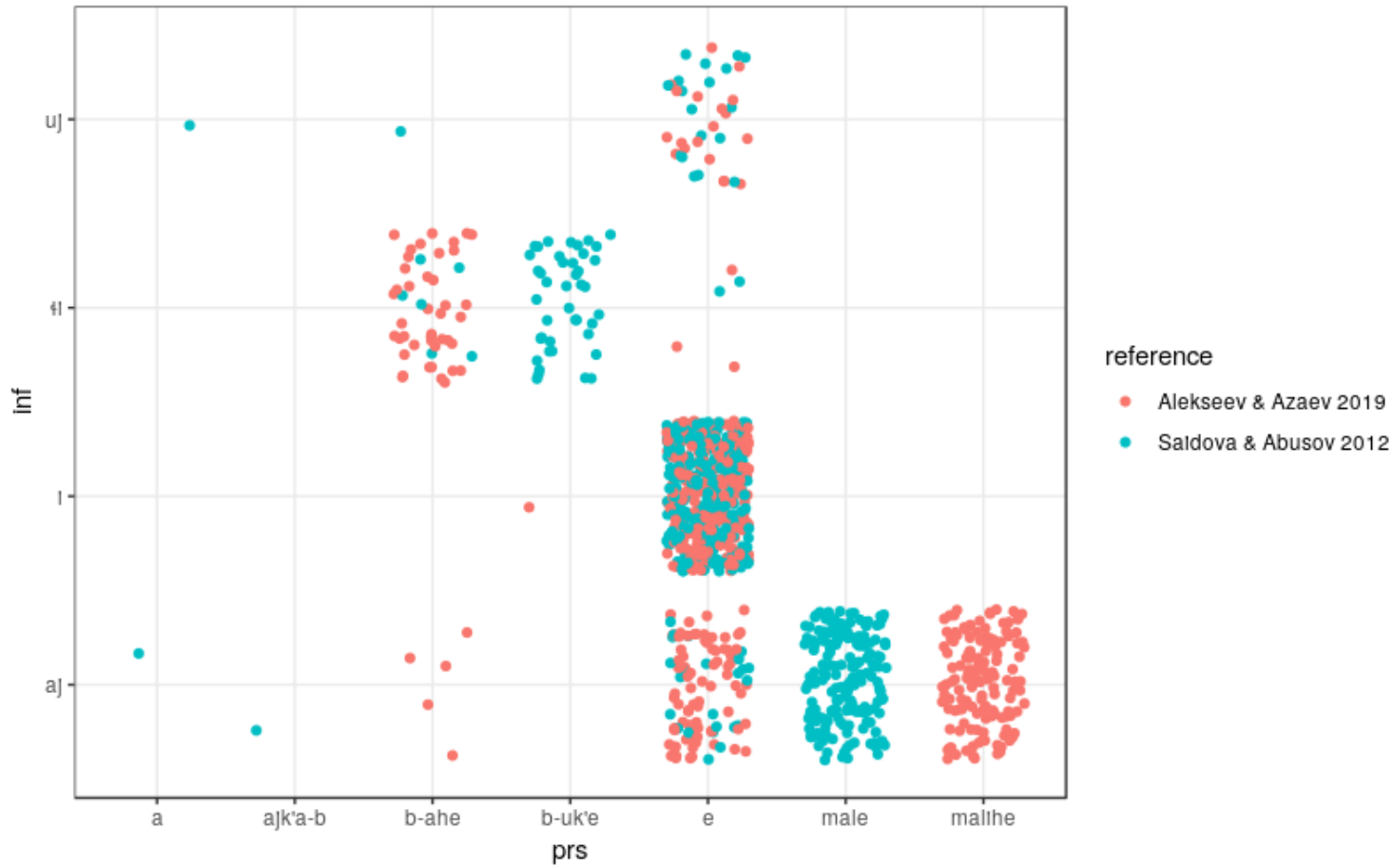
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- The bulk of material in A&A was collected several decades earlier with respect to S&A
- It seems that in contemporary Botlikh, the present form *malih-e* has been reduced to *mal-e* and the past form *malih-u* has been replaced by *-o*, formerly reconstructed as < *-*a-u* [-CAUS-PST]

Verbal morphology

Figure 6: Infinitive vs. present tense



Verbal morphology

Correspondence between infinitive and present tense form

- Illustrates the division in causative forms (cf. Table 4): full form *malihe* in A&A, shorter form *male* in S&A

Verbal morphology

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- Illustrates the division in causative forms (cf. Table 4): full form *malihe* in A&A, shorter form *male* in S&A
- Preference for a specific auxiliary to inflect derived verbs: *b-ahe* 'become' in A&A, *b-uk'a* 'be' in S&A
- This is also the case in the past

Verbal morphology

ENTRY 'BLEAT' **baʕada/ɬí**, *prs.* **bahé**, *pst.* **báhu** (Alekseev & Azaev 2019)

(1) ha-b ƛ'eť'ír **baʕáda b-uk'-é**
DEM-N lamb bleat N-be-HAB
'That lamb often bleats.'

Verbal morphology

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'That lamb often bleats.'

ENTRY 'ROAR' **bu꞉da/ɬí**, *prs.* **buké**, *pst.* **búk'a** (Saidova & Abusov 2012)

(2) únsa **bu꞉da b-áh-u**
ox roar N-become-AOR
'The ox started to roar.'

Summary

- Little phonological variation besides different stress patterns in 254 words from a 1,688 word sample
- Nouns show some variation both in the genitive and in the plural
  - A general preference for *-baɬi* over *-de* in the plural and for *-aɬi* over *-iɬi* in the genitive in S&A
  - The opposite trends in A&A
  - Variation frequently involves loanwords
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  - But more research is needed!

Summary

- Verbs show diachronic and idiosyncratic variation
  - The auxiliary used to derive the present tense of causatives has undergone morphologization: *malihe* — *male*
  - Some causatives (can) have a regular suffix *-e*
  - The past tense of the causative is formed with a suffix *-o*, which replaced an auxiliary *malihu*
  - It remains unclear how *-o* emerged

Summary

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  - Each dictionary prefers a different auxiliary as citation form for the inflection of *-ɬi* forms (derived verbs): ‘become’ in A&A, ‘be’ in S&A
  - This is a matter of personal taste: examples in the entries of these verbs show that both are possible
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  - Limited variation in other parts of the verbal domain

Спасибо



Abbreviations

AOR	aorist 39, 40
DEM	demonstrative 39, 40
HAB	habitual 39, 40
N	neuter 39, 40

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